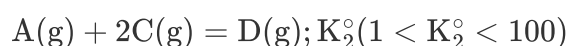
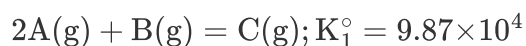


Question

At temperature T and pressure p° , reactants are charged into a long tube. On the left side, 3 mol of A and 1 mol of B are added, while on the right side, 3 mol of D and 1 mol of He are introduced. The gases on both sides are separated by a freely movable partition (initially positioned at 0 cm, impermeable to gases), and the right side also has a freely movable piston (initially positioned at 4 cm). The bottom of the tube (the far left end) is marked at -4 cm. The following reactions are known to occur:



When the reactions reach equilibrium, the piston does not move. Determine the position of the partition at equilibrium and the amount of substance of B on the left side, selecting the closest option.

A. -4 cm; 1.2×10^{-3}

B. -3.7 cm; 1.5×10^{-4}

C. -3.3 cm; ; 5.0×10^{-3}

D. -2.9 cm; 8.0×10^{-4}

E. -2.7 cm; 1.4×10^{-5}

F. -2.1 cm; 2.1×10^{-4}

G. -1.8 cm; 2.1×10^{-5}

H. -1.2 cm; 8.6×10^{-3}

I. -0.8 cm; 1.8×10^{-4}

J. -0.4 cm; 6.6×10^{-5}

K. 0.5 cm; 4.2×10^{-5}

L. 1 cm; 1.7×10^{-2}

M. 1.4 cm; 2.9×10^{-6}

N. 1.8 cm; 3.5×10^{-4}

O. 2.4 cm; 9.8×10^{-6}

P. 2.7 cm; 9.0×10^{-3}

Q. 3.2 cm; 5.1×10^{-5}

R. 3.5 cm; 1.8×10^{-6}

S. 4.0 cm; 6.9×10^{-2}

Answer

Correct Answer: **E**

Detailed Explanation

Since K_1° is much greater than K_2° , it can be assumed that the first reaction proceeds to completion, and $n(B) = 0$.

CHECKPOINT

1 PTS

$$n(B) = 0$$

Let the amount of $n(D)$ at equilibrium on the left side be x mol, the amount of $n(A)$ at equilibrium on the right side be y mol, and the position of the partition at equilibrium be a cm.

On the left side: $n(A) = (1-x)$ mol, $n(B) = 0$, $n(C) = (1-2x)$ mol, $n(D) = x$ mol;

On the right side: $n(A) = y$ mol, $n(B) = 0$, $n(C) = 2y$ mol, $n(D) = (3-y)$ mol.

$$2 - 2x + 4 + 2y = 8, y = 1 + x$$

CHECKPOINT

1 PTS

$$y = 1 + x$$

$$K_2^\circ = \frac{x(2-2x)^2}{(1-x)(1-2x)^2} = \frac{(3-y)(4+2y)^2}{4y^3}.$$

Substituting $y = 1 + x$, we solve for $x = 0.332$, and $K_2^\circ = 7.84$.

CHECKPOINT

1 PTS

$$x = 0.332$$

CHECKPOINT

1 PTS

$$K_2^\circ = 7.84$$

$$K_1^\circ = \frac{(1-x)(1-2x)^2}{(1-x)^2 n(B)_1} = \frac{2y(4+2y)^2}{y^2 n(B)_2}.$$

Solving these, we obtain $n(B)_1 = 1.36 \times 10^{-5}(\text{mol})$, $n(B)_2 = 6.76 \times 10^{-4}(\text{mol})$.

CHECKPOINT

2 PTS

$$n(B)_1 = 1.36 \times 10^{-5}(\text{mol}), n(B)_2 = 6.76 \times 10^{-4}(\text{mol})$$

Both $n(B)_1$ and $n(B)_2$ are very small, indicating that the approximation is reasonable.

$$\frac{2-2x}{4} = \frac{a+4}{4}, a = -2.66$$

CHECKPOINT

1 PTS

$$a = -2.66$$